Badge Overlay

Update Manual

Version 2026.05.13.001

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Badge Overlay is a floating badge that names the UI element under your cursor in any app, on any button, in real time. Designed for low-vision users, screen-reader supplementers, accessibility auditors, and anyone who wants a second set of eyes on what they are about to click. Works alongside YoloMouse, custom cursor schemes, and third-party cursor hooks.

1 What's New in v2026.05.13.001

v1.0.0 stable release with full WCAG 2.2 AAA conformance pass and distribution polish.

1.1 Accessibility Improvements

* **Target Size (SC 2.5.5)** - CheckBox/RadioButton min-height 36px raised to 44px. HotkeysWindow capture/clear/enable buttons 44px.
* **Focus Appearance (SC 2.4.11)** - ComboBox/ComboBoxItem now show AAA FocusRingBrush (Cyan dark/Amber light) 2px stroke + drop-shadow halo.
* **Help (SC 3.3.5)** - SectionHelpButton (44x44px click target) on Identity row; opens keyboard shortcut cheat-sheet. Also F1.
* **Animation (SC 2.3.3)** - Theme cross-fade honours both SystemParameters.ClientAreaAnimation AND per-app ReduceMotion pref.
* **Info and Relationships (SC 1.3.1)** - Radio groups wrapped in labelled GroupBoxes; screen readers announce group name before each radio.
* **Status Messages (SC 4.1.3)** - Stats 'Active for' and 'Total detections' carry LiveSetting=Polite.

1.2 Bug Fixes

* **_previousBadgeHidden cache leak** - Cache captured once on SR-Mode entry; sub-checkbox flips no longer overwrite it.
* **ToolTip / HelpText double-read** - Three controls now have distinct short/long text; eliminates redundant NVDA/JAWS double-reads.
* **Speech mute state leak** - OverlayWindow.ResetMuteState() called on SR-Mode toggle; stale mute states cannot survive across sessions.
* **EnsureAnnouncer race** - Double-checked locking around SAPI synthesizer prevents handle leaks under concurrent DPI-change recompose.
* **Idle-dim suppresses dwell-fire** - When badge is dimmed by cursor idle, dwell-mode speech no longer fires.
* **Stats 'Average rate'** - Active time now tracked separately; long-paused sessions no longer report misleadingly low detection rates.
* **Audit export target** - Live foreground-window preview every 200ms at countdown; captured handle used at countdown end.

1.3 Distribution

* SECURITY.md, CONTRIBUTING.md, THIRD-PARTY-NOTICES.txt included in installer; removed cleanly on uninstall.
* NSIS welcome screen lists headline features; installer requires licence acceptance checkbox.
* Privacy section in About redesigned as explicit blue-accent panel listing all data NOT collected.
* 'Third-Party Notices' button added to About dialog links row.

2 System Requirements

Requirement	Minimum	Recommended
OS	Windows 10 (1809+) x64	Windows 11 x64
.NET runtime	None - self-contained	None - self-contained
RAM	48 MB	128 MB+
Display	1024x600	1920x1080+
A/T	Optional	NVDA 2024+, JAWS 2024+, or Narrator

Note: No .NET runtime installation required. The executable is fully self-contained.

3 Update Instructions

3.1 Installer Update

* Close **Badge Overlay** if running (check system tray).
* Download **BadgeOverlay_Setup.exe** from the release page.
* Run installer. Detects existing installation and upgrades in-place.
* All settings, presets, and preferences are preserved.
* On first launch a **What's New** dialog highlights changes in this version.

3.2 Portable Update

* Download **BadgeOverlay_Portable_<version>.zip**.
* Extract to a temporary folder.
* Copy **BadgeOverlay.exe** over the existing portable executable.
* Your **Data** folder (settings, presets, logs) is untouched.

3.3 Silent Update

```
BadgeOverlay_Setup.exe /S
```

Installs or updates silently. Suitable for managed deployment.

3.4 Rollback

Run a previous version's installer to downgrade. Settings are preserved. Versioned builds are in the **Builds** subfolder of the project.

4 Feature Overview

Feature	Detail
Always-on labelling	Compact pill follows cursor; names button, icon, tab, link, or control beneath it.

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Universal app support	Win32, WPF, WinForms, UWP, Office ribbon, Chrome, Edge, Firefox, Electron, File Explorer.
Info Overlay	Optional large-text side panel (12pt-72pt); fully keyboard-draggable.
Custom rules	Teach the detector your own button names by control class or visible text.
Category colours	8 semantic categories with live WCAG AAA contrast verification.
Global hotkeys	Toggle, badge freeze, copy current label; fully remappable.
Screen Reader Mode	SAPI TTS announces detected labels; dwell-mode fire on pause optional.
High-contrast support	HighContrastMonitor detects SystemEvents.UserPreferenceChanged; swaps theme.
Reduce-motion	Animations skipped when ClientAreaAnimation or ReduceMotion pref is off.
5 named presets	Store appearance settings; Ctrl+1-5 to load; Ctrl+Shift+1-5 to save.
Crash-safe settings	Write-to-temp-then-rename prevents corrupt prefs.json on power loss.
F1 cheat-sheet	All keyboard shortcuts in one window, generated from current bindings.

5 Accessibility (WCAG 2.2 AAA)

This application targets WCAG 2.2 Level AAA across all interactive surfaces.

Criterion	Implementation
Contrast (SC 1.4.6 AAA 7:1)	All text tokens verified >= 7:1 on Dark and Light themes.
Non-text contrast (SC 1.4.11)	Focus rings (Cyan/Gold) >= 3:1; slider stroke + drop-shadow halo.
Touch targets (SC 2.5.5 AAA 44px)	All primary nav and action buttons >= 44x44px.
Focus appearance (SC 2.4.11 AAA)	2px Gold (light) / Cyan (dark) focus ring on every interactive control.
No colour-only info (SC 1.4.1)	Status uses glyphs alongside colour coding.
Live regions (SC 4.1.3)	AutomationProperties.LiveSetting=Polite on stats controls.
High-contrast mode	HighContrastMonitor swaps to system-colour theme on demand.
Reduce-motion (SC 2.3.3)	ClientAreaAnimation and ReduceMotion pref both respected.
Keyboard navigation (SC 2.1.1)	Arrow-key drag on Info Overlay; logical Tab order; F1 shortcut ref.
Authentication (SC 3.3.8)	None required. No accounts. No cognitive tests.

6 Known Issues & Notes

Compact dialog density

Settings panels use 32-36px button heights in dense groups vs the AAA-recommended 44px. Primary nav and action buttons are 44px. Documented intentional deviation.

Resize handle - keyboard

Bottom-right resize triangle is mouse-only. Use Win+Arrow window snapping. Keyboard resize planned.

Per-application profiles

Per-app automatic preset switching is planned. Use manual Ctrl+1-5 preset loading in the interim.

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This document is provided for informational purposes. Specifications subject to change.